Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
6	*(a)*		**Vertical**: Decreasing (accept deceleration), then increasing (accept acceleration) / <u>changes</u> at 9.81 m s^{-2} (1) **Horizontal**: Constant (1) **Reason**: Gravity acts vertically **or** no forces act horizontally (1)	3			3		
	(b)	(i)	0.15 [m s^{-1}] i.e. $\frac{(1000)}{(110 \times 60)}$ or 0.54 km/h		1		1		
		(ii)	Correct substitution into $x = ut + ½ at^2$ Ignore sign convention e.g. 1 000 = ½ × a × (55 × 60)^2 (1) At least one mathematical step shown leading to a = 0.00018 [m s^{-2}] e.g. $a = \frac{2\,000}{1.09 \times 10^7}$ (1) **Alternative:** u_{vertical} calculated from $x = ½(u + v)t$ i.e. u = 0.606 m s^{-1} (1) Substitution into: $a = (v-u)/t$ to show a = 0.00018 [m s^{-2}] (1)	1	1		2	2	
		(iii)	Correct substitution into $v = u + at$ **or** $v^2 = u^2 + 2ax$ e.g. 0 = u – 0.00018 (55 × 60) **or** 0 = u^2 – 2 × 0.00018 × 1 000 (1) **ecf** [accept use of 0.0002 m s^{-2}] e.g. u = 0.61 [m s^{-1}] (1) e.g. $\frac{0.61}{0.88} \times 100\%$ seen (1) Accept 67% – 75% **Alternative for final mark:** 60% of 0.88 m s^{-1} = 0.53 m s^{-1} therefore: 0.61 > 0.53	1	1 1		3	3	